

Discrimination and Economic Development*

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Abstract

Following the abolition of slavery, a new set of extractive institutions emerged in the US South collectively known as Jim Crow. This paper studies both the endogenous formation of these institutions and their consequences for economic development. We propose a unified empirical framework that traces the chain from growing Black populations to the adoption of restrictive institutions to their impact on long-run development. Using a shift-share IV design based on within-South Black migration, we show that places with faster Black population growth adopted harsher anti-Black institutions, and that those institutions slowed local economic development while benefiting white elites. Black migration shocks translated into both local anti-Black violence and statewide political responses that held back development across the South. The quantitative evidence aligns with the long-standing hypothesis that land-owning elites imposed anti-Black institutions on Southern society for their own gain, and it explains why the South only started to catch up with non-Southern economic development after the end of Jim Crow.

JEL Codes: J15, N31, N41, O43, P16

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1. INTRODUCTION

The American South has been poorer than the rest of the United States for more than 150 years. Different factors have contributed to the region’s underdevelopment: specialization in agriculture (North, 1961), the destruction wrought by the Civil War (Engerman, 1966; Goldin and Lewis, 1975; Feigenbaum et al., 2022), and the lasting economic consequences of slavery (Dell, 2010; Fujiwara et al., 2023). The South was also the epicenter of Jim Crow—a set of anti-Black institutions of racial segregation, Black disenfranchisement, and restrictions to Black economic and geographic mobility (Du Bois, 1935; Key, 1949; Wright, 1986; Ransom and Sutch, 2001).

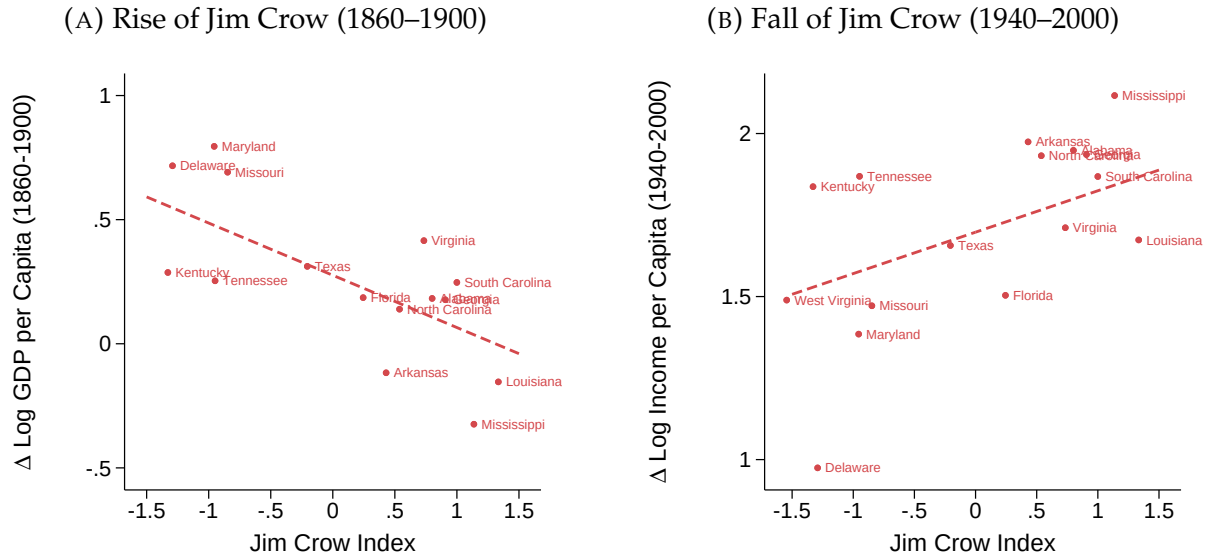
In this paper, we study the role of legal and extralegal racial discrimination in the South’s long-run underdevelopment. We propose a unified empirical framework that traces the chain from growing Black populations to violence and the adoption of restrictive institutions to their impact on economic development. Using a shift-share IV design based on within-South Black migration, we show that places with faster Black population growth adopted harsher Jim Crow institutions—and that those institutions slowed economic development while benefiting white elites.

We begin by establishing a set of descriptive facts. First, the South’s economic gap with the rest of the country widened after the Civil War and only began to close in the mid-twentieth century. Second, during the Jim Crow era, the gap grew most in states that adopted the harshest Jim Crow laws; in contrast, the gap shrank most in the same states when those laws were dismantled during the Civil Rights era (see Figure 1). The widening gap during Jim Crow and its reversal during Civil Rights establish a negative correlation between Jim Crow intensity and economic development. Third, states with large and growing Black populations were more likely to adopt restrictive legislation, and counties experiencing faster Black population growth were more likely to vote for the Democratic party—which led the Jim Crow agenda. Where Black populations grew, white-dominated legislatures responded with repressive institutions, and those institutions held back economic development.

The mechanism linking Black population growth to repression runs through both elite self-interest and a cross-class white coalition. White elites had a direct economic interest in preserving access to cheap Black labor, while many poorer whites faced labor-market competition from Black workers.

One of [the planters] said: “We have white men working on the farms. We frequently have applications every day. But when the white men come and are willing to work we have to say: We cannot afford to pay you any more, because I can get a negro for 60¢ a day; if you are willing to work at that price the first vacancy we have you can have it. We occasionally put a white man in that way.” (Wright, 1987, p. 104)

FIGURE 1: Jim Crow and Economic Development Over Time



Notes: Panel A plots the relationship between state Jim Crow intensity and the change in GDP per capita between 1860 and 1900, using GDP data from [Francis \(2025\)](#). Panel B plots the relationship between state Jim Crow intensity and the change in income per capita between 1940 and 2000, using state personal income data from the Bureau of Economic Analysis ([U.S. Bureau of Economic Analysis, 2026](#)). Each bubble is a state, with bubble size proportional to population.

A bargain was needed to preserve the coalition between elites and workers, who were the voting bloc of the Democratic party. The resulting institutional framework protected elite rents while securing broader white support through legally and socially enforced status distinctions—including what [Du Bois \(1935\)](#) calls a “psychological wage,” the social and legal privileges that compensated poorer whites for lowered economic progress.

The theory of laboring class unity [...] would throw white and black labor into one class [but] this failed to work in the South [...] because the theory of race was supplemented by a carefully planned and slowly evolved method which drove such a wedge between the white and black workers [...] The white group of laborers [...] were compensated in part by a sort of public and psychological wage. They were given public deference and titles of courtesy because they were white [...] The result of this was that the wages of both classes could be kept low. ([Du Bois, 1935](#), p. 688-689)

Because state legislatures enacted the most consequential restrictions, local migration shocks translated into statewide political and institutional responses. To identify the effect of Black population growth on anti-Black institutions and economic development, we propose a new shift-share IV design based on within-South Black migration. Our empirical strategy uses historical county linkages and distant origin-specific push and pull factors to isolate plausibly exogenous changes in local Black population growth after 1870 and to trace their long-run consequences through 2000. Our strategy generalizes several aspects of the shift-share design of [Boustan \(2010\)](#) and [Derenoncourt \(2022\)](#) to a within-South setting in which counties are both origins and destinations and migration

is ongoing rather than concentrated in a single episode.

Our first main result is that places exposed to larger increases in Black population growth became more supportive of anti-Black politics and saw stronger adoption of Jim Crow laws. A one-decile increase in Black population change raises Democratic vote share by roughly four percentage points in presidential elections and six percentage points in gubernatorial elections. The voting effect is robust to controlling for local economic conditions and region-level fixed effects. The same variation predicts significantly more Jim Crow laws targeting employment, migration, and suffrage—the categories most directly tied to labor control and political exclusion. All results are robust to controlling for white migration and to excluding adjacent states from the instrument; placebo regressions using pre-period (1855–1856) political outcomes show only a weak relationship with the later migration instrument.

Our second main result is that the primary channel runs through state-level institutions, not local demographic composition alone. Once we control for aggregate state-level Black population growth, Jim Crow institutions, or state fixed effects, the local migration coefficients fall by more than half—consistent with the fact that state legislatures, not county governments, enacted the most consequential Jim Crow legislation.

Our third main result is that the resulting institutional regime slowed economic development while benefiting white elites. Counties with instrumented increases in Black population rank lower on the agricultural and industrial development sub-indices during the decades when states built Jim Crow. Mechanization lagged and educational inequality widened. Black Americans in high-migration counties experienced the largest economic losses. In contrast, farm values rose, consistent with a labor-intensive agricultural path that raised land values while constraining structural transformation. The pattern of broad development costs alongside concentrated elite gains complements [Althoff and Reichardt \(2024\)](#), who use a border-based research design following individual Black and white families from their 1870 origin location to show that Jim Crow reduced Black economic progress and raised white elite incomes.

The editor of the Macon, Georgia, Telegraph said recently: "Since at least 1865, we have been holding back the Negro to keep him from getting beyond the white man. Our idea has been that the Negro should be kept poor. But by keeping him poor, we have thrown him into competition with ourselves and have kept ourselves poor. Of course, Governor Talmadge has the popular attitude. It is to hold the Negro down in order to make him work -to keep him poor. And Southerners are willing to keep themselves and their kind and section down and poor in order to keep the Negro that way." ([Du Bois, 1935](#), p. 684)

This paper provides long-run causal evidence that racial discrimination slowed Southern economic development. The South's postbellum experience has long been understood as a low-wage, labor-repressive equilibrium ([Wright, 1986, 2013](#)), and recent work shows that dismantling Jim Crow through Civil-Rights-era enforcement increased Black

political participation, redirected public resources, and generated broad economic gains (Cascio and Washington, 2014; Ang, 2019). Our results explain the other side of this pattern: the institutions that the Civil Rights Movement dismantled had depressed Southern development for decades, which helps explain both why the South fell further behind after emancipation and why sustained convergence began only after Jim Crow ended.

This paper identifies a unifying mechanism linking the endogenous adoption of discriminatory institutions to economic underdevelopment. A large literature documents the persistent effects of colonial extractive institutions on development (Acemoglu et al., 2001, 2002; Dell, 2010) and the power of elites to impede change even after formal regime transitions (Sokoloff and Engerman, 2000; Acemoglu and Robinson, 2008). Jim Crow illustrates this dynamic: abolition and Reconstruction changed the formal political order, but white elites retained enough de facto power to erect a new extractive regime. Our results show how Black population growth triggered this institutional response and how the resulting violence and laws depressed development while protecting elite rents.

“The planters had to move toward the control of the political power of newly enfranchised labor... The movement of the planter class to attract black labor... met the immediate and bitter fear and opposition of the poor whites... These desperately feared the rise of black labor to a position which might equal and even surpass the poor whites... The poor whites long fought to exclude the Negro in apportioning the political power of the state, and after Reconstruction united in disfranchising him... all the time... was the feeling that unless the planters united with the poor whites in a solid racial phalanx against the black voters, anarchy and destruction were preferable to the economic rise of the Negro.” (Du Bois, 1935, p. 599 - 601)

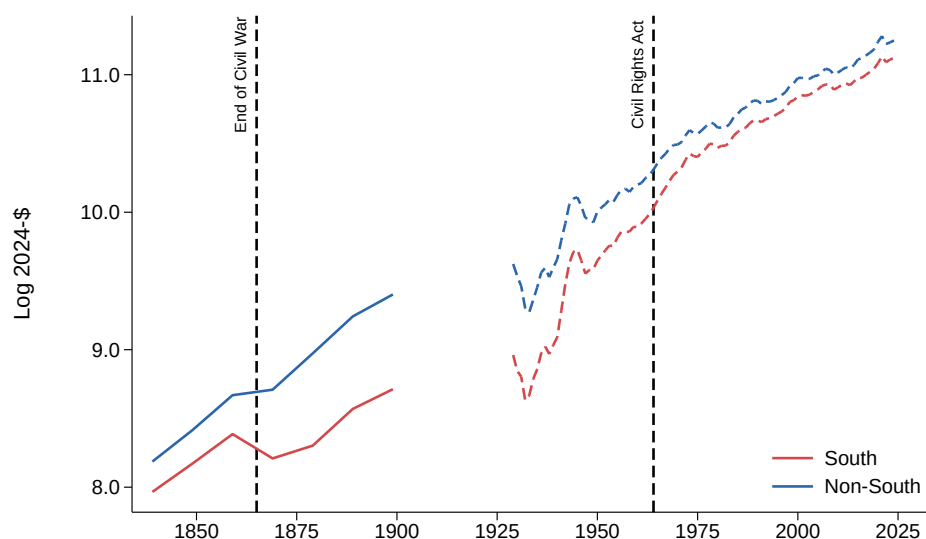
We broaden our understanding of how Jim Crow affected economic outcomes to the joint incidence on aggregate development and elite rents. Existing research shows that coercive postbellum institutions constrained Black mobility, wages, and long-run economic progress (Naidu, 2010; Althoff and Reichardt, 2024), and that Black demographic and political gains triggered white backlash (Logan, 2020, 2023; Chyn et al., 2024; Anderson et al., 2025). We study the political formation of these institutions and trace their consequences for the broader Southern economy. Jim Crow reduced local development while concentrating gains among white elites—a pattern that helps explain why the regime endured for nearly a century despite its wider economic costs.

2. HISTORICAL BACKGROUND

The US South’s persistent underdevelopment has gone hand in hand with a changing set of racially discriminatory institutions. Slavery ended in 1865 but was followed by Jim Crow—an almost century-long period in which states disenfranchised Black voters, restricted Black geographic and economic mobility, segregated most aspects of daily life,

provided public goods unequally, and tolerated extralegal violence against Black Americans. Jim Crow only ended with the Civil Rights Movement in the 1950s and 1960s.

FIGURE 2: The South's Persistent Economic Lag



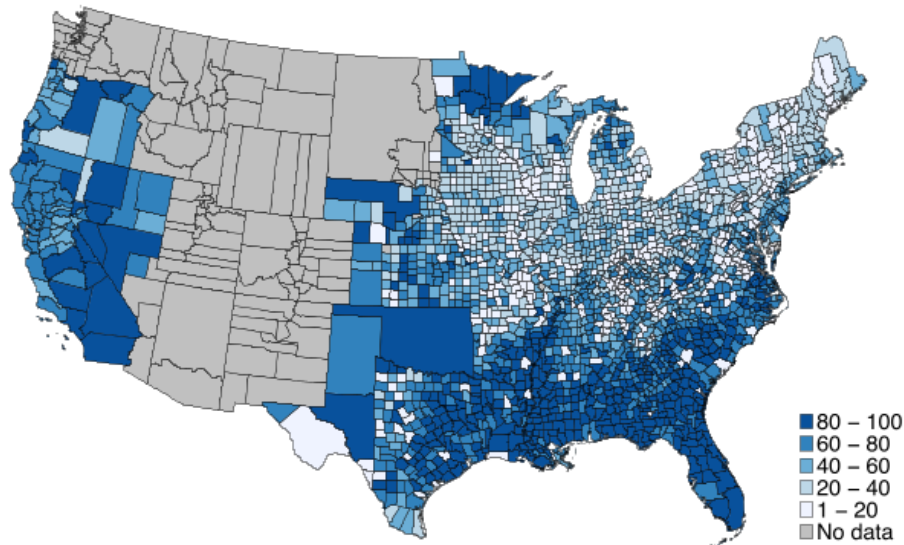
Notes: This figure shows GDP per capita for the South and non-South from 1840 to 2024. GDP per capita for 1840–1900 is from [Francis \(2025\)](#). State personal income per capita for 1929–2024 is from the Bureau of Economic Analysis ([U.S. Bureau of Economic Analysis, 2026](#)).

Each Southern state had its own Jim Crow regime, varying in intensity and scope, with the most severe restrictions emerging in states with the largest Black populations. The concentration of Black populations shifted after the end of slavery. Black Americans left plantations, reunited families, searched for safer communities, bargained over wages, and pursued land and autonomy.¹ Many moved westward toward Arkansas, Louisiana, and Mississippi, where labor demand and the prospect of economic independence were strongest before the rise of Jim Crow ([Ransom and Sutch, 2001](#); [Wright, 1986](#)).

In 1866, there was a sufficient migration of Negroes from Georgia to the West to cause some alarm. ([Du Bois, 1935](#), p. 489)

¹Indeed, even some Black men from the North moved South during Reconstruction: “...numerous northern [Black Americans] who moved south after the end of slavery, one of the few times in American history when the region offered [B]lack men of talent and ambition greater opportunities than the North.” ([Foner, 2006](#)).

FIGURE 3: Ranked Changes in Black Population (1870–1900)



Notes: This map shows county-level percentile ranks of the change in the Black population share between 1870 and 1900. Darker counties experienced larger increases in Black population share. Data are from linked full-count census records.

Large and growing Black populations threatened white elites' grip on Southern politics and the Southern economy. In counties where Black residents approached or exceeded a majority, unrestricted suffrage could overturn the planter class's hold on state legislatures and local government (Key, 1949). At the same time, emancipation gave workers the ability to leave, eroding landowners' bargaining power and making the plantation order harder to sustain. The early response included coercive and improvised measures like restrictive labor contracts, debt peonage, vagrancy statutes, limits on recruiters, and violence, which all raised the cost of exit (Ransom and Sutch, 2001; Naidu, 2010). Over time these coercive labor-market restrictions were codified into law. Jim Crow combined disenfranchisement, segregation, restrictions on economic and geographic mobility, and unequal provision of public goods (Du Bois, 1935; Key, 1949; Althoff and Reichardt, 2024). The result was often a reversal of fortunes: Black workers moved to high-productivity plantation areas immediately after emancipation, only to face increasingly oppressive institutions in those same places (Du Bois, 1935; Key, 1949; Wright, 1986). Because the key rules were written at the state level, migration shocks in one part of a state could reshape institutions far beyond the counties that directly received migrants.

Sustaining Jim Crow, however, required not only elite self-interest but also a cross-class white coalition. White elites wanted a dependable low-wage labor force; poorer whites, who faced direct labor-market competition from Black workers, wanted protection of a different kind. The regime served both by preserving elite access to cheap Black labor while granting ordinary whites legal privileges, electoral dominance, and the status

benefits that Du Bois calls a “psychological wage” (Du Bois, 1935).

White planter elites did not vanish after defeat; they adapted, retaining leverage over land and labor through sharecropping and related arrangements after Congress abandoned Radical promises of land redistribution (Wiener, 1978). Fragile Black political rights lacked a durable economic base, and Jim Crow emerged as an elite-led effort to re-anchor control over Black labor and votes.

It must be remembered and never forgotten that the civil war in the South which overthrew Reconstruction was a determined effort to reduce black labor as nearly as possible to a condition of unlimited exploitation and build a new class of capitalists on this foundation. The wage of the Negro worker, despite the war amendments, was to be reduced to the level of bare subsistence by taxation, peonage, caste, and every method of discrimination. This program had to be carried out in open defiance of the clear letter of the law. The lawlessness in the South since the Civil War has varied in its phases. First, it was that kind of disregard for law which follows all war. Then it became a labor war, an attempt on the part of impoverished capitalists and landholders to force laborers to work on the capitalist’s own terms. From this, it changed to a war between laborers, white and black men fighting for the same jobs. Afterward, the white laborer joined the white landholder and capitalist and beat the black laborer into subjection through secret organizations and the rise of a new doctrine of race hatred. (Du Bois, 1935, p. 659)

The substantial within-South reallocation of Black population during this period—driven by the search for land, wages, and autonomy—provides the source of variation we exploit in the next section to identify the causal effect of Black population growth on institutions and development.

3. DATA

We combine county-level demographic, economic, and political data from 1850 to 2000. The identifying variation comes from county-to-county migration flows measured in linked full-count census records. We aggregate county-level outcome data from the Census of Agriculture and the Census of Manufacturing, full-count census microdata for individual-level socioeconomic outcomes, election returns from ICPSR, a Jim Crow law database, and records of lynchings and Ku Klux Klan activity.

Migration networks. We construct county-to-county migration shares from linked full-count IPUMS census records (Ruggles et al., 2024) using the crosswalk files from the Census Linking Project (Abramitzky et al., 2020). For each pair of consecutive census decades from 1850–1860 through 1930–1940 (excluding 1890, which was lost in a fire), we link individuals across censuses and compute the share of an origin county’s population

that moved to each destination county. These (lagged) shares form the basis of the shift-share instrument.

Population. County-level population counts for white and Black individuals from 1850 to 1940 come from the *Historical, Demographic, Economic and Social Data: The United States, 1790–2002* (ICPSR 2896) (Haines, 2010). The population counts provide the denominators for population-change measures and per-capita scaling.

County-level agricultural and economic outcomes. Agricultural outcomes—including farm values, acreage, crop production, machinery, tenure composition, and labor inputs—come from the NHGIS-harmonized county-level Census of Agriculture panel (Manson et al., 2021) (replacing the original ICPSR 35206 series), covering 1840–1950. Manufacturing outcomes—including establishments, capital, employment, wages, and output—come from ICPSR 37114 (1929–1935) and ICPSR 38834 (1946–1974). We harmonize all outcomes to consistent 1870 county boundaries using a population-weighted crosswalk.

Individual-level census outcomes. We use full-count IPUMS census microdata for 1850–1940 (Ruggles et al., 2024) to construct county-level measures of school attendance, literacy, occupational income scores, wage income (available from 1940), manufacturing employment, and self-employment, separately by race. The individual-level measures are aggregated to 1870-boundary counties and used to study schooling gaps, income differences, and labor-market outcomes that are not available in published county aggregates.

Election data. County-level presidential and gubernatorial election returns come from ICPSR 00001 (Inter-university Consortium for Political and Social Research, 1999). We construct a balanced panel of Democratic vote shares by decade from 1870 to 1960 for 15 Southern states. Details on interpolation and state-specific adjustments are in Appendix D.

Jim Crow laws. We use the state-level Jim Crow law database and Jim Crow index from Althoff and Reichardt (2024). The database records discriminatory laws by state, year, and type (employment, migration, suffrage, education, transportation, marriage, and other). We construct counts of discriminatory laws enacted by 1900 and by type for the analysis of institutional responses.

Lynchings. County-level lynching counts come from two sources: the national lynching dataset of Seguin and Rigby (2019), covering 1883–1941, and the Equal Justice Initiative (EJI) county supplement (Equal Justice Initiative, 2024).

Ku Klux Klan. Data on the second Ku Klux Klan (1920s) come from the Virginia Commonwealth University Libraries’ mapping project (Kneebone and Torres, 2015), which records individual klavern locations. We aggregate these to county-level counts and construct an indicator for any KKK presence, crosswalked to 1870 county boundaries.

State legislative voting. We are assembling a new dataset on state legislators to study who supported Jim Crow in practice. We are digitizing roll-call votes on Jim Crow bills

from state legislative journals. Data collection is ongoing: we have complete data for Georgia, Louisiana, North Carolina, Texas, and Virginia, with collection in progress for Alabama and Kentucky, and the data include suffrage restrictions, labor and employment laws, and segregation statutes. We link legislators to full-count census records using name, age, and state of residence, yielding legislators' counties of residence and linking legislative behavior to local migration exposure. These data are not yet used in the main-text results. Table 22 in Appendix D summarizes current state coverage.

Linking Legislators to the Census. We link the Texas legislators to the full-count census for 1860, 1870, 1880, and 1900 to describe their socioeconomic position. We standardize names using the cleaning routine of [Abramitzky et al. \(2020\)](#), restrict candidate records to men in the legislator's county of residence or counties of representation, and score the similarity between the legislator's name and each candidate's name with the Jaro-Winkler metric on first and last names. We retain candidates with a similarity above 0.80 and assign each legislator the highest-scoring record. Candidate records are restricted to white men aged 17 and older, except during Reconstruction (1868–1876) and for legislators identified as Black, where records of all races are retained.

We begin with 2,294 individuals with the name and county information. We link between 16 and 20 percent of them to each census: 377 in 1860 (16.4 percent), 436 in 1870 (19.0 percent), 454 in 1880 (19.8 percent), and 371 in 1900 (16.2 percent). The denominator includes every legislator regardless of whether they were alive and resident in Texas in a given census year, so these shares understate the rate at which we recover legislators who were present in the state at the time of enumeration.

To describe how matched legislators compare to the broader population, we use as a reference group white male household heads aged 17 and older in the same counties. Matched legislators are older and report more property. In 1860, the median matched legislator reported \$5,700 in total property (in nominal census-year dollars), against \$1,300 for other household heads; in 1870 the corresponding medians are \$2,534 and \$600. Based on linked slave-schedule records, 46 percent of matched legislators in 1860 owned slaves, against 22 percent of other household heads. Matched legislators also held higher-status occupations: in 1880, their median occupational income score was 30, against 14 for other household heads. Occupationally, matched legislators are concentrated among farm owners, professionals such as lawyers and physicians, and merchants and proprietors, and they include few laborers relative to the household-head population.

4. EMPIRICAL STRATEGY

The key empirical challenge is selection into migration: Black migrants may have moved toward places with different economic or political conditions. Our strategy exploits the

fact that Black migrants followed established settlement networks: we predict where they moved using historical migration patterns and agricultural push factors in their counties of origin. Because the predicted migration is driven by conditions in other counties rather than in the destination, it isolates variation in local Black population growth that is plausibly unrelated to destination-specific conditions.

4.1. Shift-Share IV Design

The goal is to estimate the causal effect of Black in-migration on institutions and development, accounting for the fact that Black migrants may move toward places with better economic opportunities or different political conditions. We therefore adapt the Black migration shift-share design of [Boustan \(2010\)](#) and [Derenoncourt \(2022\)](#) to a within-South setting in which counties are both origins and destinations and migration is ongoing rather than concentrated in a single episode. The design uses historical migration-network shares to map shocks from origin counties into destination counties, so identification does not require classifying places in advance as fixed origins or destinations.

4.2. Instrument Construction

We define the treatment as the cumulative change in the Black population of destination county d between 1870 and end year T , normalized by the county's 1870 total population:

$$\Delta b_d^{1870-T} = \frac{\text{Black pop}_d^T - \text{Black pop}_d^{1870}}{\text{Total pop}_d^{1870}}. \quad (1)$$

The main specification uses $T = 1900$, which captures the migration shock during Jim Crow's formative period. We then use this migration exposure to study downstream political, institutional, and economic consequences over the long run, including outcomes through 2000.

The instrument predicts cumulative Black population change using two ingredients: historical cross-county migration shares and predicted outmigration from other counties based on lagged agricultural push factors. For each target decade $t \in \{1880, 1890, \dots, T\}$, the decade-specific predicted change in county d is

$$\widehat{\Delta b}_{d,t} = \sum_{o \neq d} \text{share}_{o \rightarrow d}^{t-20, t-10} \cdot \widehat{\text{out-mig}}_o^{t-10, t}, \quad (2)$$

where $\text{share}_{o \rightarrow d}^{t-20, t-10}$ is the fraction of white migrants from origin o who settled in destination d , computed from linked census records in the decade ending ten years before the target; and $\widehat{\text{out-mig}}_o^{t-10, t}$ is predicted Black outmigration from county o during the

decade from $t - 10$ to t , estimated from lagged agricultural push factors using LASSO for variable selection and leave-one-state-out prediction. Shares are based on white linked flows rather than Black flows to avoid a mechanical correlation between the instrument and the endogenous variable.

The cumulative instrument for the window 1870– T sums these decade-specific components:

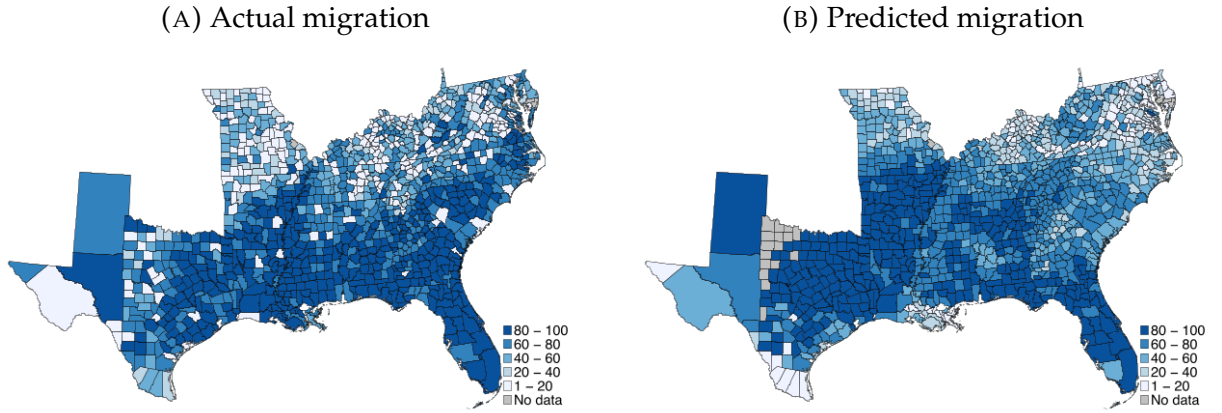
$$\widehat{\Delta b}_d^{1870-T} = \sum_{t=1880}^T \widehat{\Delta b}_{d,t}. \quad (3)$$

Both the treatment and the instrument are normalized by the county's 1870 total population. In our preferred specification, we convert the cumulative treatment and instrument to percentile ranks (rescaled to a 1–10 decile scale), so that

$$R(\Delta b_d) = \text{rank}(\Delta b_d^{1870-T}), \quad R(\widehat{\Delta b}_d) = \text{rank}(\widehat{\Delta b}_d^{1870-T}). \quad (4)$$

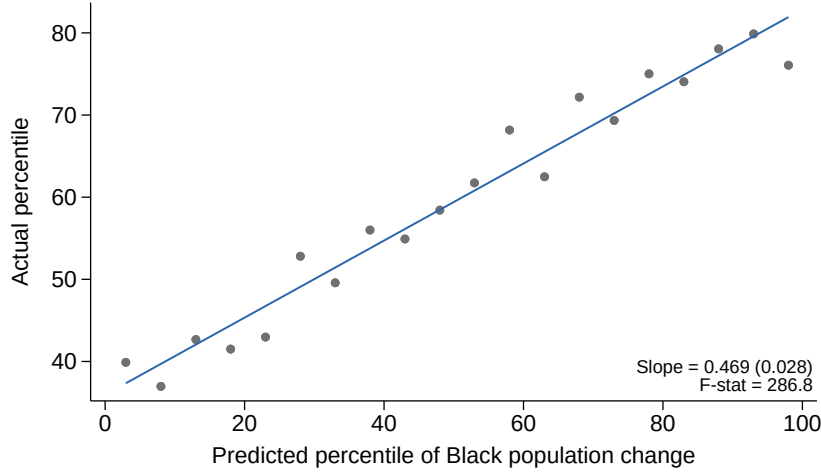
This ranked specification is robust to outliers and ensures that coefficients reflect the effect of moving one decile in the distribution of predicted Black population change, rather than depending on the scale of level changes.

FIGURE 4: Actual vs. Predicted Migration



Notes: Panel A maps county-level percentile ranks of the actual change in the Black population share between 1870 and 1900. Panel B maps the corresponding predicted change from the shift-share instrument. Darker shading indicates larger increases. Data are from linked full-count census records.

FIGURE 5: First Stage: Predicted vs. Actual Black Population Change



Notes: This figure plots binned averages of actual Black population change against predicted Black population change across Southern counties between 1870 and 1900. Each marker represents a bin of counties, and the line shows the linear fit.

4.3. Identification and Validation

Identification requires that, conditional on destination-level controls, the instrument shifts local Black population growth because of shocks in other counties rather than because of omitted pull factors in destination county d . We construct predicted inflows into d from leave-one-state-out origin shocks and historical migration links, so the identifying variation comes from conditions outside the destination county.

The key assumption is that predicted migration to destination county d is driven by push factors in origin county $o \neq d$, not pull factors in destination county d :

$$\mathbb{E}[R(\widehat{\Delta b}_d) \times \varepsilon_d \mid \mathbf{X}'_d] = 0 \quad (5)$$

conditional on destination-level controls $\mathbf{X}'_d$.

We assess the exclusion-restriction assumption in several ways. First, placebo regressions show that the instrument does not predict pre-period political outcomes in 1855–1856. Second, we reconstruct the instrument while excluding adjacent states, which focuses attention on longer-distance migration flows. Third, we control for white migration to absorb broader population movements that could be correlated with local conditions. Finally, following [Borusyak et al. \(2022\)](#), we control for incomplete shares to separate variation from origin shocks from variation in the density of migration links at the destination.

5. RESULTS: POLITICAL OUTCOMES

5.1. Voting Patterns

Counties that received larger Black inflows voted substantially more Democratic. In the specification with controls and region fixed effects, moving one decile higher in Black population change raises the Democratic presidential vote share by 3.8 percentage points and the gubernatorial vote share by 5.9 percentage points (Table 1). Relative to sample means of 62 and 68 percent, moving from the bottom to the top decile of Black in-migration would raise Democratic support by roughly 34 and 53 percentage points, respectively—large enough to shift a competitive county into a one-party stronghold. The raw correlation between Black population change and Democratic vote share is visible in Appendix Figure 6.

A natural concern is that the estimates reflect state-level confounds rather than local demographic pressure. When we control for aggregate state-level Black migration, the Jim Crow index, or state fixed effects, the coefficients attenuate substantially: for presidential elections, the coefficient falls to near zero with state-level migration controls and to about 1.7 with state fixed effects; for gubernatorial elections, the corresponding estimates are 1.5 and 2.3—less than half of the original effect. Rather than undermining the main finding, the attenuation confirms that the political response operated primarily through state-level channels: state legislatures, not county governments, enacted the most consequential Jim Crow legislation. The results are robust to controlling for white migration, excluding adjacent states from the instrument, and placebo tests using 1855–1856 political outcomes (Appendix A).

TABLE 1: Democratic Vote Share

	Democratic vote share in 1900 (%)									
	Presidential					Gubernatorial				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Decile of Black population change	3.83*** (0.33)	3.85*** (0.36)	4.03*** (0.38)	0.88 (0.63)	2.82*** (0.44)	5.18*** (0.37)	5.42*** (0.41)	5.95*** (0.45)	2.48*** (0.65)	4.64*** (0.51)
Controls	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Region FE	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
State-level total Black migration	No	No	No	Yes	No	No	No	No	Yes	No
Jim Crow index	No	No	No	No	Yes	No	No	No	No	Yes
First-stage F-stat	165.77	187.96	178.84	53.52	124.35	165.63	187.45	178.28	53.20	123.89
Observations	1,146	1,146	1,146	1,146	1,146	1,148	1,148	1,148	1,148	1,148
Sample mean	60.71					65.94				

Notes: This table reports IV estimates of the effect of Black population change on Democratic vote share in presidential and gubernatorial elections in 1900. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.2. Jim Crow Laws

Black in-migration also predicts the adoption of formal anti-Black institutions. The strongest effects appear for the categories most directly tied to labor control and political exclusion: a one-decile increase in Black population change predicts 1.1 additional employment laws, 0.5 additional migration laws, and 0.25 additional suffrage restrictions (Table 2). The estimates for transportation and marriage laws are imprecise, while the education and residual “other” categories are negative. The pattern across categories points to a targeted institutional response rather than a uniform increase across every type of statute.

TABLE 2: Jim Crow Laws by Type

	Employment	Migration	Suffrage	Education	Transportation	Marriage	Other
Decile of Black population change	1.14*** (0.33)	0.59*** (0.10)	0.26** (0.13)	-0.46 (0.29)	0.49 (0.32)	-0.01 (0.36)	-0.27 (0.17)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	No	No	No	No	No	No	No
First-stage F-stat	184.68	184.68	184.68	184.68	184.68	184.68	184.68
State clusters	17	17	17	17	17	17	17

Notes: This table reports IV estimates of the effect of Black population change on the number of Jim Crow laws adopted by category. The instrument is the predicted Black population change from the shift-share design. Standard errors are clustered at the state level and are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5.3. Extralegal Violence

Counties with larger Black inflows were more likely to have a KKK presence: a one-decile increase in Black population change raises the probability of KKK presence by 7.7 percentage points in the specification with controls and region fixed effects (Table 3). In contrast to the voting results, the KKK estimates are not attenuated by state-level factors, suggesting that formal legislation and extralegal violence operated independently rather than as alternatives. The lynching estimates differ across the EJI and historical series and the coefficients change sign or lose significance once additional controls or state-level political variables are added. The same local pattern appears for sundown towns: a one-decile increase in Black population change raises the sundown-town rate by about 8 towns per million white residents in 1900, robust to controls and region fixed effects (Appendix Table 15, col. 3); the binary indicator for any sundown town in a county attenuates to insignificance once controls are added, suggesting the migration response operated through how aggressively counties adopted such ordinances rather than through the extensive margin of which counties adopted at all. Outside the South, [Bazzi et al. \(2022\)](#) link sundown-town adoption to Southern White out-migration as part of the “Great Retreat”; our estimates show a parallel within-South response to local Black in-migration.

Violence on the part of the Ku Klux Klan [sought] to subject black labor to strict domination by capital and to break Negro political power [...] It did disfranchise black labor with the aid of white Southern labor and with the silent acquiescence of white Northern labor. (Du Bois, 1935, p. 617)

TABLE 3: KKK Presence

	Any KKK clan present (p.p.)				
	(1)	(2)	(3)	(4)	(5)
Decile of Black population change	9.31*** (1.11)	8.21*** (1.14)	7.73*** (1.17)	12.60*** (2.34)	9.13*** (1.43)
Controls	No	Yes	Yes	Yes	Yes
Region FE	No	No	Yes	Yes	Yes
State-level total Black migration	No	No	No	Yes	No
Jim Crow index	No	No	No	No	Yes
Cragg-Donald F-stat	164.83	184.68	175.60	51.03	121.95
Observations	1,154	1,154	1,154	1,154	1,153

Notes: This table reports IV estimates of the effect of Black population change on county-level KKK presence. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 4: Historical Lynchings

	Number of lynchings 1882-1930 (per 1M Black residents in 1900)				
	(1)	(2)	(3)	(4)	(5)
Decile of Black population change	400.60** (166.51)	139.30 (134.37)	363.17** (156.13)	742.28 (533.00)	471.94* (247.32)
Controls	No	Yes	Yes	Yes	Yes
Region FE	No	No	Yes	Yes	Yes
State-level total Black migration	No	No	No	Yes	No
Jim Crow index	No	No	No	No	Yes
First-stage F-stat	164.83	184.68	175.60	51.03	121.95
Observations	1,154	1,154	1,154	1,154	1,153

Notes: This table reports IV estimates of the effect of Black population change on county-level lynching rates, using both EJI and historical records. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6. RESULTS: ECONOMIC DEVELOPMENT

Counties exposed to larger Black in-migration experienced weaker economic development during the decades when states built Jim Crow, with the penalty concentrated in the between-state institutional channel. In the specification with region fixed effects, high-migration counties rank lower on the agricultural development sub-index throughout the period (and on the industrial sub-index in the early decades), show slower mechanization, and have lower income in 1940. Farm values and output per acre, however, rise—consistent with a labor-intensive agricultural path that raised land values while constraining structural transformation; indeed the agricultural sub-index, which penalizes the associated shift toward tenancy and sharecropping, is precisely the measure that captures this structural stagnation. Once state fixed effects absorb the institutional channel, the local migration coefficients attenuate or reverse, reinforcing the interpretation that state-level Jim Crow legislation drove the development penalty. We first examine the development sub-indices across decades, then turn to specific outcomes: agricultural structure, mechanization, and wage income in 1940.

6.1. Economic Development Index

To characterize the development trajectory along conceptually distinct margins, we construct three county-level development sub-indices for each decade from 1880 to 1950: an *agricultural* index (farm capital, mechanization, output and value per acre, livestock, and the structure of land tenure), an *industrial* index (manufacturing establishments, capital, output, wages, and horsepower; urbanization and population density; and railroad market access), and a *human-capital* index (school attendance, literacy, occupational standing, years of schooling, and patenting). Each sub-index is the average of its available component indicators for that decade, standardized and signed so that higher values denote greater development—indicators of agrarian backwardness, such as tenancy and sharecropping shares, enter negatively—and then converted to a percentile rank (0–100). Because the set of available indicators varies across decades, each sub-index is constructed separately for each decade and scope and is left blank in decades with too few indicators (the industrial index cannot be built for 1890). We also construct a “white” human-capital variant that restricts the schooling, literacy, and occupation inputs to white residents; the agricultural and industrial indices are place-level measures for which the historical censuses provide no racial breakdown.

Splitting development into three conceptually distinct margins shows that Black in-migration bore on them in different directions (Table 5). In the specification with controls and region fixed effects, *agricultural* development is significantly lower in every decade from 1880 to 1950: counties exposed to larger Black inflows shifted toward a tenant-

and sharecropper-based, labor-intensive agriculture that is structurally less modern even where land values rose. This structural agrarian penalty is the clearest and most persistent development cost. *Industrial* development is significantly lower in 1880 but turns significantly *higher* from 1900 onward, and *human-capital* development is mixed early and turns positive by 1940–1950.

TABLE 5: Development Sub-Indices by Decade: Agricultural, Industrial, and Human Capital

	1880	1890	1900	1910	1920	1930	1940	1950
<i>Panel A. Agricultural development</i>								
Decile of	-3.01*	-7.27***	-2.51***	-4.19***	-4.08***	-5.03***	-4.77***	-5.06***
Black population change	(1.74)	(1.05)	(0.66)	(0.66)	(0.66)	(0.57)	(0.55)	(0.54)
<i>Panel B. Industrial development</i>								
Decile of	-8.63***		2.39***	2.59***	1.74***	3.69***	3.04***	2.85***
Black population change	(1.72)		(0.63)	(0.59)	(0.58)	(0.56)	(0.52)	(0.55)
<i>Panel C. Human capital</i>								
Decile of	-2.66*	1.65**	-0.13	-0.03	0.74	0.77	2.35***	3.03***
Black population change	(1.38)	(0.79)	(0.57)	(0.57)	(0.57)	(0.58)	(0.54)	(0.57)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F	28.18	103.88	175.37	185.53	180.54	193.44	223.73	201.13
N	1151	1152	1151	1152	1152	1152	1152	1152

Notes: This table reports IV estimates of the effect of Black population change (in deciles) on three county-level development sub-indices—agricultural (Panel A), industrial (Panel B), and human capital (Panel C)—each expressed as a 0–100 percentile rank, by decade. Each sub-index is the average of its available standardized component indicators for that decade, signed so that higher values denote greater development (indicators of agrarian backwardness such as tenancy and sharecropping enter negatively); it is not a principal component. The industrial index is not defined for 1890 because the requisite manufacturing indicators are unavailable that decade; the cell is left blank. All specifications include controls and region fixed effects. First-stage *F* and the number of observations are reported once at the foot; the first stage is common across sub-indices. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 6: Development Sub-Indices by Decade: Black and White Population Change

	1880	1890	1900	1910	1920	1930	1940	1950
<i>Panel A. Agricultural development</i>								
Decile of Black population change	3.61 (6.55)	-10.24*** (2.34)	-5.16*** (1.37)	-7.93*** (1.72)	-8.49*** (1.49)	-9.38*** (1.14)	-11.13*** (1.15)	-12.12*** (1.27)
Decile of White population change	-19.37 (21.68)	5.45 (3.85)	4.65** (2.18)	6.36** (2.76)	6.79*** (2.14)	6.22*** (1.31)	9.29*** (1.31)	10.09*** (1.38)
<i>Panel B. Industrial development</i>								
Decile of Black population change	-4.74 (5.44)		1.05 (1.29)	4.29*** (1.50)	1.24 (1.18)	2.39** (1.04)	-0.22 (0.94)	2.30** (1.10)
Decile of White population change	-11.37 (17.94)		1.35 (2.13)	-3.78 (2.52)	0.82 (1.70)	1.38 (1.19)	4.51*** (1.04)	0.82 (1.12)
<i>Panel C. Human capital</i>								
Decile of Black population change	-0.04 (3.26)	-4.87** (2.09)	-6.72*** (1.28)	-6.50*** (1.36)	-4.82*** (1.08)	-4.43*** (1.09)	-2.18** (0.91)	-1.22 (1.07)
Decile of White population change	-3.61 (11.07)	11.89*** (3.43)	11.51*** (2.17)	10.86*** (2.42)	8.32*** (1.71)	6.77*** (1.36)	6.60*** (1.07)	5.71*** (1.17)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F	0.25	6.58	12.72	9.89	16.76	27.20	37.42	30.08
N	1151	1152	1151	1152	1152	1152	1152	1152

Notes: This table repeats Table 5 while separately instrumenting Black and white population change (each in deciles). Within each panel the first row reports the effect of the decile of Black population change and the second the effect of the decile of white population change. Sub-indices, sample, and controls are as in Table 5, and the industrial index is undefined for 1890. The joint (Black and white) first stage is weak in the earliest decades—the reported first-stage F is well below conventional thresholds in 1880 (and, for industrial, 1890)—so those columns should be read with caution. First-stage F and the number of observations are reported once at the foot. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.2. Income Effects

The 1940 income results point to a broad regional penalty associated with places exposed to larger Black inflows. In specifications with controls and region fixed effects, larger Black population change predicts lower overall income, lower white income, and especially lower Black income. Once state-level migration or the Jim Crow index is added, the overall and white coefficients move close to zero, while Black income remains negative. With state fixed effects, the overall and white coefficients turn positive and the Black coefficient becomes close to zero. The attenuation under state fixed effects again points to state-level institutions as the main margin through which Black in-migration affected later income outcomes.

TABLE 7: Income (1940)

	(1)	(2)	(3)	(4)
Decile of Black population change	-0.02*** (0.01)	-0.02*** (0.01)	0.01 (0.01)	-0.01 (0.01)
Controls	Yes	Yes	Yes	Yes
Region FE	No	Yes	Yes	Yes
State migration aggregate	No	No	Yes	No
Jim Crow index	No	No	No	Yes
Cragg-Donald F-stat	182.81	173.54	50.23	121.69
Observations	1152	1152	1152	1151

Notes: This table reports IV estimates of the effect of Black population change on county-level log average income in 1940. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 8: White Income (1940)

	(1)	(2)	(3)	(4)
Decile of Black population change	-0.02*** (0.01)	-0.02** (0.01)	0.01 (0.01)	-0.00 (0.01)
Controls	Yes	Yes	Yes	Yes
Region FE	No	Yes	Yes	Yes
State migration aggregate	No	No	Yes	No
Jim Crow index	No	No	No	Yes
Cragg-Donald F-stat	182.81	173.54	50.23	121.69
Observations	1152	1152	1152	1151

Notes: This table reports IV estimates of the effect of Black population change on county-level log white income in 1940. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 9: Black Income (1940)

	(1)	(2)	(3)	(4)
Decile of Black population change	-0.06*** (0.01)	-0.05*** (0.01)	-0.03** (0.02)	-0.05*** (0.01)
Controls	Yes	Yes	Yes	Yes
Region FE	No	Yes	Yes	Yes
State migration aggregate	No	No	Yes	No
Jim Crow index	No	No	No	Yes
Cragg-Donald F-stat	169.10	167.97	49.15	118.38
Observations	1118	1118	1118	1117

Notes: This table reports IV estimates of the effect of Black population change on county-level log Black income in 1940. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Manufacturing activity grew in counties that received Black migrants: both manufacturing establishments per capita and especially manufacturing capital per capita increase. This may point to the complementarity between cotton production and cotton processing, although we have not yet explicitly tested this hypothesis. With state fixed effects, the wage-income coefficients for the overall and white samples turn positive, while manufacturing capital remains positive.

TABLE 10: Income and Manufacturing (1870–1900)

	Wage income (log)			Manufacturing estab.	Manufacturing capital
	All	White	Black	per capita (log)	per capita (log)
Decile of Black population change	-0.02*** (0.01)	-0.02** (0.01)	-0.05*** (0.01)	-0.00 (0.00)	0.22*** (0.04)
Controls	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
First-stage F	173.54	173.54	167.97	174.53	174.53
N	1152	1152	1118	1152	1152

Notes: This table reports IV estimates of the effect of Black population change on county-level wage income and manufacturing outcomes between 1870 and 1900. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.3. Farms and Agricultural Mechanization

Counties with larger Black inflows see higher agricultural output and farm values but no increase in mechanization, consistent with landowners extracting more value from labor rather than investing in capital (Table 11). Agricultural output per acre rises by

0.054 log points and total farm value per acre by 0.087 log points, both statistically significant. Farm machinery per acre is unchanged—the coefficient is close to zero and insignificant—and farm size does not respond. The pattern points to a labor-intensive agricultural path: additional Black workers raised land productivity and land values while giving landowners little incentive to mechanize.

TABLE 11: Agricultural Development (1870–1900)

	Total farm value per capita (log)	Farm machinery per capita (log)	Livestock value per capita (log)
Decile of Black population change	0.07*** (0.02)	0.04 (0.03)	-0.09*** (0.01)
Controls	Yes	Yes	Yes
Region FE	Yes	Yes	Yes
First-stage F	175.27	175.27	175.27
N	1149	1149	1149

Notes: This table reports IV estimates of the effect of Black population change on county-level agricultural development outcomes between 1870 and 1900. Outcomes include farm structure, output, asset values, and mechanization measures. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6.4. Heterogeneity by Occupational Segregation

The political backlash is somewhat stronger in counties with low initial occupational segregation: Democratic vote shares rise by 4.0 and 5.9 percentage points below the median, compared with 3.3 and 5.3 above (Table 12). White workers responded more sharply to Black in-migration where the two groups competed in the same occupations. Farm values per capita also increase more in low-segregation counties—0.074 versus 0.045 log points—as do machinery values, which are marginally significant below the median but insignificant above. The 1940 income penalty, by contrast, is concentrated in high-segregation counties: a 2.3 percent decline, significant at the five percent level, versus a small and insignificant effect below the median.

TABLE 12: Heterogeneity by Initial Occupational Segregation

	Occupational segregation in 1870	
	Below median	Above median
Democratic presidential vote share in 1900 (%)	4.19*** (0.54)	3.44*** (0.60)
Democratic gubernatorial vote share in 1900 (%)	6.20*** (0.66)	5.49*** (0.69)
Total farm value per capita (1870–1900 log change)	0.07** (0.03)	0.04 (0.03)
Machinery value per capita (1870–1900 log change)	0.07** (0.03)	0.04 (0.03)
Average wage income in 1940 (log)	-0.02 (0.01)	-0.03** (0.01)
Segregation mean (D-index)	0.02	0.19
First-stage F	73.40	58.24
N (median sample)	509	510

Notes: This table reports IV estimates of the effect of Black population change, split by below- and above-median initial occupational segregation (D-index) in 1870. Outcomes include voting, agricultural development, mechanization, and 1940 income. All specifications include baseline controls and region fixed effects. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7. CONCLUSION

This paper links the South’s long-run underdevelopment to the political response to Black mobility after emancipation. Counties exposed to larger Black in-migration shifted toward anti-Black voting and were located in states that adopted more restrictive Jim Crow laws. The estimates consistently point to state-level politics as the main channel through which local demographic change translated into durable institutions.

Jim Crow did not simply redistribute political rights. It also shaped economic development. The county-level evidence points to weaker development across a range of outcomes in the states where Jim Crow became strongest, even though some effects vary across specifications and outcomes. Together with prior evidence that Jim Crow benefited politically powerful whites, the results suggest a broader mechanism: discriminatory institutions can persist because they generate concentrated gains for powerful groups while reducing aggregate development. Institutions that served elites but slowed growth help explain both the South’s postbellum divergence and its later convergence after Jim Crow was dismantled.

An important limitation is that our measures of Jim Crow capture formal statutes but not the full apparatus of racial control. Informal enforcement—extralegal violence, social pressure, employer collusion, and selective application of vagrancy laws—likely rein-

forced and sometimes substituted for formal institutions, but informal enforcement is difficult to measure systematically. The KKK and lynching results provide partial evidence on extralegal enforcement, yet they almost certainly understate the scope of informal coercion. Future work linking the legislative voting data we are assembling to local migration exposure can identify how demographic pressure translated into specific Jim Crow statutes.

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Appendices

A. ROBUSTNESS: POLITICAL OUTCOMES

TABLE 13: Effect of Black Population Change on Democratic Vote Share (White Migration Control)

	Democratic vote share in 1900 (%)									
	Presidential					Gubernatorial				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Decile of Black population change	3.83*** (0.33)	6.81*** (0.84)	7.79*** (1.12)	4.00*** (1.18)	6.36*** (1.12)	5.18*** (0.37)	8.65*** (0.95)	9.40*** (1.23)	4.96*** (1.17)	7.74*** (1.22)
Decile of White population change		-4.73*** (1.18)	-6.95*** (1.85)	-5.15*** (1.44)	-5.94*** (1.69)		-5.20*** (1.35)	-6.48*** (2.05)	-4.39*** (1.51)	-5.32*** (1.85)
Controls	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Region FE	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
State-level total Black migration	No	No	No	Yes	No	No	No	No	Yes	No
Jim Crow index	No	No	No	No	Yes	No	No	No	No	Yes
First-stage F-stat	165.77	24.24	12.48	10.92	11.94	165.63	24.29	12.57	10.98	12.01
Observations	1,146	1,146	1,146	1,146	1,146	1,148	1,148	1,148	1,148	1,148
Sample mean			60.71					65.94		

Notes: This table replicates Table 1 while additionally controlling for white population change between 1870 and 1900. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE 14: Effect of Black Population Change on Democratic Vote Share (Dropping Adjacent States)

	Democratic vote share in 1900 (%)									
	Presidential					Gubernatorial				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Decile of Black population change	3.83*** (0.33)	3.85*** (0.36)	4.03*** (0.38)	0.88 (0.63)	2.82*** (0.44)	5.18*** (0.37)	5.42*** (0.41)	5.95*** (0.45)	2.48*** (0.65)	4.64*** (0.51)
Controls	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Region FE	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
State-level total Black migration	No	No	No	Yes	No	No	No	No	Yes	No
Jim Crow index	No	No	No	No	Yes	No	No	No	No	Yes
First-stage F-stat	165.77	187.96	178.84	53.52	124.35	165.63	187.45	178.28	53.20	123.89
Observations	1,146	1,146	1,146	1,146	1,146	1,148	1,148	1,148	1,148	1,148
Sample mean			60.71					65.94		

Notes: This table replicates Table 1 using an instrument that excludes adjacent states when constructing predicted Black migration. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B. EXTRALEGAL VIOLENCE: SUNDOWN TOWNS

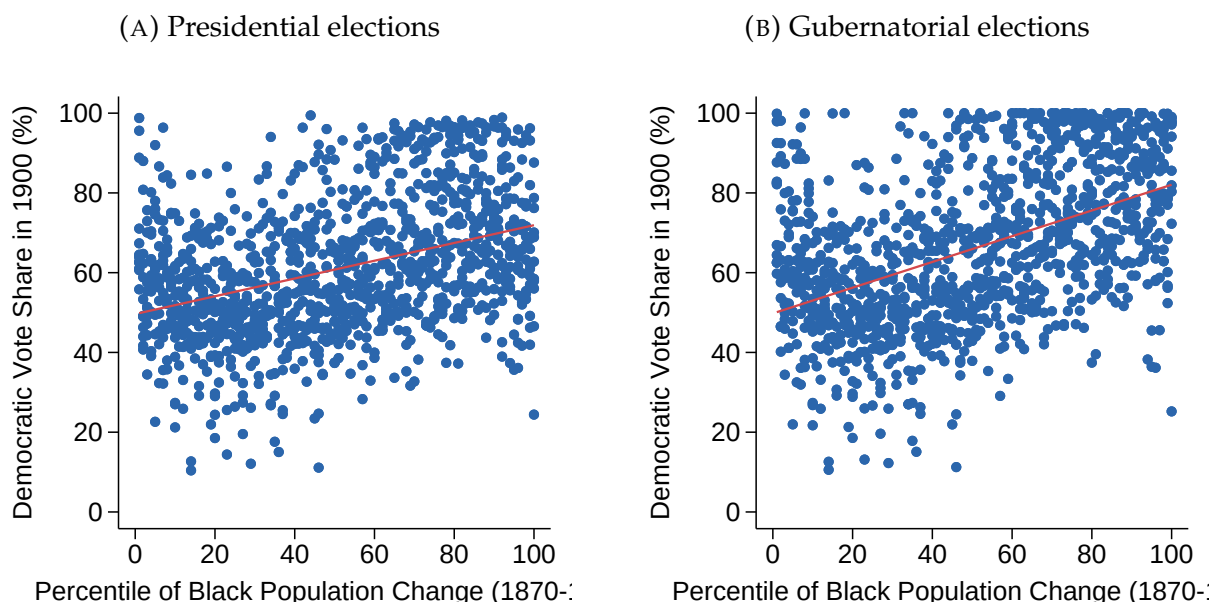
TABLE 15: Sundown Towns

	Any sundown town present					Sundown towns per 1M white residents in 1900				
	(1)	(2)	(3)	(4)	(5)	(1)	(2)	(3)	(4)	(5)
Decile of Black population change	0.02** (0.01)	0.00 (0.01)	0.01 (0.01)	-0.00 (0.02)	0.00 (0.01)	1.09e+01*** (4.04e+00)	7.34e+00** (2.90e+00)	6.29e+00** (2.85e+00)	2.61e+00 (5.99e+00)	8.12e+00* (4.67e+00)
Controls	No	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes
Region FE	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
State-level total Black migration	No	No	No	Yes	No	No	No	No	Yes	No
Jim Crow index	No	No	No	No	Yes	No	No	No	No	Yes
Cragg-Donald F-stat	164.83	184.68	175.60	51.03	121.95	164.83	184.68	175.60	51.03	121.95
Observations	1,154	1,154	1,154	1,154	1,153	1,154	1,154	1,154	1,154	1,153

Notes: This table reports IV estimates of the effect of Black population change (in deciles) on sundown-town outcomes at the 1870-county level. The left panel uses an indicator for any confirmed sundown town in the county; the right panel uses the count of confirmed sundown towns per one million white residents in 1900. Sundown-town records are from [Rigby and Seguin \(2025\)](#), assigned to 1870 counties by point-in-polygon on each town's coordinates. The instrument is the predicted Black population change from the shift-share design. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

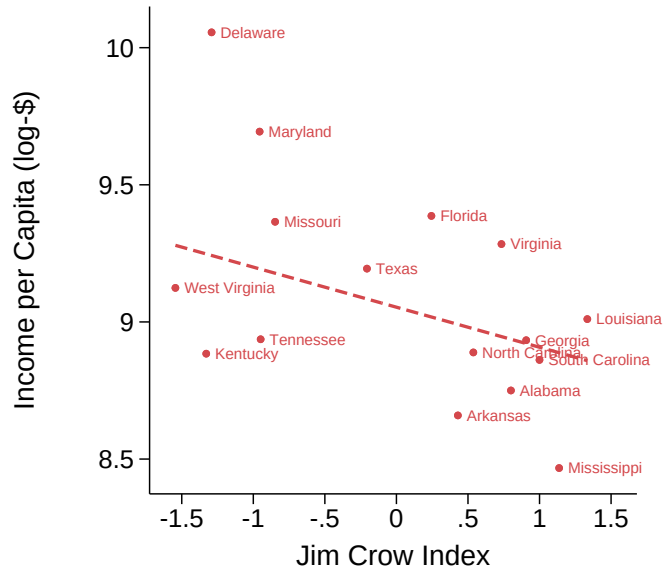
C. ADDITIONAL FIGURES

FIGURE 6: Black In-Migration and Political Outcomes



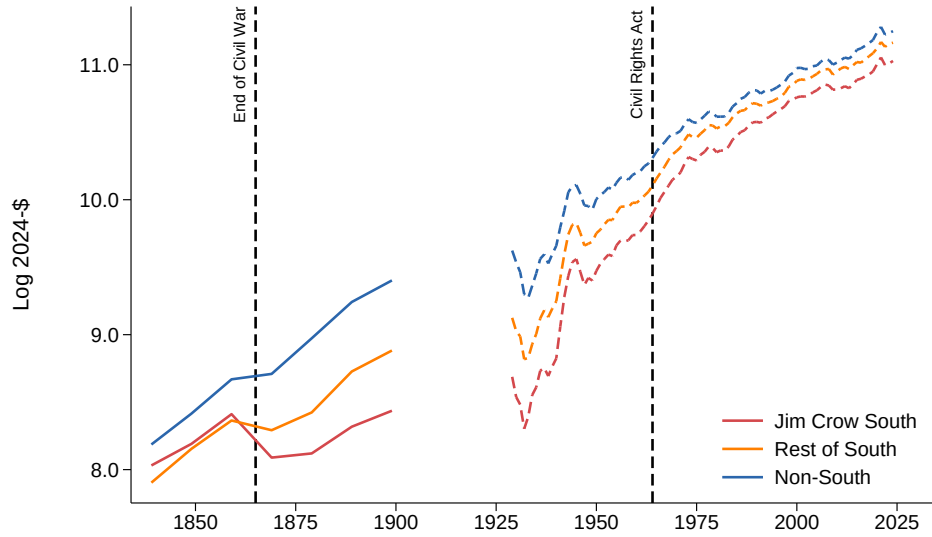
Notes: This figure shows the raw correlation between the change in Black population share between 1870 and 1900 and Democratic vote share in 1900 across Southern counties. Panel A uses presidential elections. Panel B uses gubernatorial elections.

FIGURE 7: Jim Crow and State Income Levels (1940)



Notes: Each bubble is a state, with bubble size proportional to population. The x-axis plots the Jim Crow index from [Althoff and Reichardt \(2024\)](#), and the y-axis plots state personal income per capita in 1940 from the Bureau of Economic Analysis ([U.S. Bureau of Economic Analysis, 2026](#)).

FIGURE 8: Jim Crow and Economic Development Over Time (By Sub-Region)



Notes: This figure plots GDP per capita for the Upper South, Lower South, and non-South from 1840 to 2024. GDP per capita for 1840–1900 is from [Francis \(2025\)](#). State personal income per capita for 1929–2024 is from the Bureau of Economic Analysis ([U.S. Bureau of Economic Analysis, 2026](#)).

D. DATA APPENDIX

We describe the construction of each variable used in the analysis.

D.1. Migration Shares

We use linked historical US census data (1850–1940) from full-count IPUMS records ([Ruggles et al., 2024](#)) to measure migration shares used in the shift-share instrument, together with the Census Linking Project crosswalk files ([Abramitzky et al., 2020](#)). These migration shares represent the fraction of individuals from an origin county who migrated to specific destination counties between census decades. We examine all census decade pairs (1850–1860, 1850–1870, . . . , 1930–1940), excluding the 1890 census (lost in a fire). We create historical migration shares for both white and Black Americans.

D.2. Population

We use detailed county- and state-level descriptive data from the *Historical, Demographic, Economic and Social Data: The United States, 1790–2002* (ICPSR 2896) ([Haines, 2010](#)). Based on county-level population counts for white and Black individuals from 1850 to 1940, we compute county- and state-level population changes by decade.

D.3. Predictors

We use data from ICPSR 2896 ([Haines, 2010](#)), specifically the agriculture, manufacturing, and population censuses for our set of predictors. Table 16 presents the different categories of variables we consider as predictors.

TABLE 16: LASSO Candidate Predictor Categories

Category	Variables	Decades
Agriculture	Farmland area (improved, woodland, tilled, meadow, unimproved), number of farms by size category (7–8 bins), farm values and equipment, farm labor costs, farm output, orchard products, livestock counts (horses, mules, cows, oxen, cattle, sheep, swine), crop production (corn, wheat, cotton, oats, rice, tobacco, barley), farm tenure (tenants, owners)	1870–1930
Manufacturing	Number of establishments, capital (total, buildings, equipment, land), employment (total; by gender and age from 1880), wages, raw materials, output value. Per-worker ratios (skilled wages, total wages, output, capital per worker) and operation schedules from ICPSR 09385 (1880 only). Derived ratios: steam/water horsepower shares, employment per establishment, wages per worker	1870–1880, 1890–1900, 1920–1930
Fiscal	Tax revenue by level (non-national, state, county, local), county and local debt	1870–1880
Irrigation	Irrigated establishments, irrigated acres	1890–1900
$\Delta \log$	Log-differenced per capita versions of agricultural, manufacturing, and fiscal variables across consecutive census decades (e.g., $\Delta \log$ 1870–1860 for predicting 1880 migration)	All decades

Notes: This table summarizes the categories of candidate predictors from which the LASSO algorithm selects covariates for predicting county-level net migration rates. All level variables are measured in per capita terms. The number of candidates ranges from approximately 25 to 50 per decade, depending on data availability.

D.4. Shock Prediction

We create different versions of the shocks for the SSIV design: (i) baseline prediction, using LASSO (with both level and log-differenced candidates) and a leave-one-state-out method to predict migration for Black and white Americans; (ii) leave-adjacent-states-out prediction; (iii) decade-by-decade prediction; (iv) log-differenced predictors only.

D.4.1 LASSO

Under the assumption that county-level variation in Southern economic indicators from 1870–1890 is uncorrelated with Southern destination county characteristics for migrants from those counties, we view estimating Southern county net migration rates as a pure prediction problem. Following [Belloni et al. \(2010\)](#) and [Derenoncourt \(2022\)](#), we use a machine-learning-based estimation in a context where the number of potential predictors is large relative to the number of observations. We apply this approach to select predictors in the “zero” stage prediction of net migration from origin counties using push factors, based on the predictors mentioned above:

$$\text{mig rate}_{jt} = \beta_0 + \mathbf{Z}'_{jt-10}\beta + \varepsilon_{jt}, \quad (6)$$

for $t \in \{1880, 1890, \dots, 1940\}$ where mig rate_{jt} is net migration for county j from the

decade of $t - 10$ to t and $\mathbf{Z}_{jt-10}$ is the set of candidate predictors measured in $t - 10$, including both level and log-differenced variables. Using LASSO, we shrink the set of predictors to an optimal subset. Excluding a predictor from the subset corresponds to setting the respective element of β to zero. More explicitly, LASSO solves the following problem:

$$\min_{\beta_0, \beta} \left\{ \frac{1}{N} \sum_{j=1}^N \left(\text{mig rate}_{jt} - \beta_0 - \mathbf{Z}'_{jt-10} \beta \right)^2 \right\} \text{ subject to } \sum_{k=1}^K |\beta_k| \leq \lambda, \quad (7)$$

where N is the number of Southern counties and K is the total number of potential predictors, including agricultural, demographic, and manufacturing characteristics. λ is selected to minimize the Bayesian Information Criterion (BIC).

Table 16 summarizes the categories of candidate predictors available for LASSO selection (see above). Different decades have different selected variables because λ is chosen separately for each decade. In the baseline specification, the candidate pool includes both level and log-differenced predictors. Table 17 reports the covariates selected by the LASSO algorithm for each decade.

TABLE 17: LASSO-Selected Covariates for Black Migration Prediction (South)

Decade	Selected covariates
1880	Swine p.c., corn p.c., cotton p.c., mfg. capital p.c., mfg. employment per estab. (rank), Δ log cotton p.c., Δ log peas & beans p.c., Δ log 10–19 acre farms p.c., Δ log 25,000+ urban pop. p.c., Δ log mfg. estab. p.c., Δ log Baptist congregations p.c., Δ log Jewish congregations p.c., Δ log church property value p.c., Δ log male pop. p.c., Δ log mfg. emp. per estab.
1890	City/town taxes p.c., mfg. emp. per estab. (rank), Δ log tobacco p.c., Δ log 25,000+ urban pop. p.c., Δ log Chinese pop. p.c., Δ log born in AL p.c., Δ log born in GA p.c., Δ log born in MS p.c., Δ log born in SC p.c., Δ log born in France p.c., Δ log born in Ireland p.c., Δ log city/town taxes p.c.
1900	Wheat p.c., mfg. output p.c., Δ log rice p.c., Δ log tobacco p.c., Δ log cotton p.c., Δ log clover seed p.c., Δ log grass seed p.c., Δ log male mfg. workers 16+ p.c., Δ log Chinese pop. p.c., Δ log born in Russia p.c., Δ log born in Spain p.c., Δ log flax straw p.c., Δ log cane molasses p.c., Δ log rice acreage p.c., Δ log tobacco acreage p.c.
1910	Farms 3–9 acres p.c., farms 1,000+ acres p.c., mfg. establishments p.c., mfg. capital in buildings p.c., mfg. output p.c., Δ log farm values p.c., Δ log wheat p.c., Δ log 25,000+ urban pop. p.c., Δ log mfg. output p.c., Δ log female pop. p.c., Δ log male mfg. workers 16+ p.c., Δ log born in Poland p.c., Δ log born in Southeast Asia p.c., Δ log state Black pop. change since 1870 p.c., Δ log hemp acreage p.c., Δ log peanuts p.c., Δ log oranges p.c., Δ log born in Hungary p.c., Δ log born in Sweden p.c., Δ log born in Wales p.c., Δ log state Black pop. change since 1880 p.c.
1920	Livestock value p.c., rice p.c., farms 20–49 acres p.c., Δ log 25,000+ urban pop. p.c., Δ log state White pop. change since 1870 p.c., Δ log other grasses p.c., Δ log lemons p.c., Δ log olives p.c., Δ log raspberry acreage p.c.
1930	Farm value p.c., mfg. establishments p.c., mfg. raw materials p.c., mfg. output p.c., Δ log live-stock value p.c., Δ log total pop. p.c., Δ log 25,000+ urban pop. p.c., Δ log White male pop. p.c., Δ log female pop. p.c., Δ log total pop. (alt.) p.c., Δ log state Black pop. p.c., Δ log state White pop. p.c., Δ log tenant farms p.c., Δ log farm area p.c., Δ log state Black pop. change since 1870 p.c., Δ log state White pop. change since 1870 p.c., Δ log owned farms (alt.) p.c., Δ log farm buildings value p.c., Δ log farm equipment value p.c., Δ log fruit value p.c., Δ log Black illiteracy 10+ p.c., Δ log farms by foreign-born whites p.c., Δ log farm acreage p.c., Δ log owned acres p.c., Δ log improved owned acres p.c., Δ log farm value owned p.c., Δ log tenanted acres p.c., Δ log improved tenanted acres p.c., Δ log farm value tenanted p.c., Δ log farm output tenanted p.c., Δ log crop value p.c., Δ log other grain value p.c., Δ log hay value p.c., Δ log vegetable value p.c., Δ log other crop value p.c., Δ log total pop. 10+ p.c., Δ log illiteracy 10+ p.c., Δ log total pop. 18–20 p.c., Δ log born in Belgium (white) p.c., Δ log born in French Canada (white) p.c., Δ log born in Central America (white) p.c., Δ log born in Cuba (white) p.c., Δ log born in England (white) p.c., Δ log born in Finland (white) p.c., Δ log born in Holland (white) p.c., Δ log born in Italy (white) p.c., Δ log born in Mexico (white) p.c., Δ log born in Spain (white) p.c., Δ log state Black pop. change since 1900 p.c.
1940	Farms p.c., swine p.c., farms 50–99 acres p.c., mfg. labor cost p.c., mfg. wages per worker, water HP share (rank), mfg. wages per worker (rank), Δ log woodland acres p.c., Δ log state White pop. p.c., Δ log farm acres p.c., Δ log state Black pop. change since 1870 p.c.

Notes: This table lists the covariates selected by the LASSO algorithm for predicting county-level net Black migration rates in the South specification. The candidate pool includes both level and Δ log predictors jointly. For decades with many selected variables (1910–1940), counts and category summaries are shown; full variable lists are available in the replication files. All level variables are per capita. Δ log denotes the log difference between consecutive census decades. λ is chosen separately for each decade to minimize the BIC.

D.5. Economic Development Index Inputs

TABLE 18: Selected Inputs for the Agricultural Development Index by Decade

Decade	Selected index inputs (South)
1880	ag output value pa; farm value per acre pa; farms owner share; farms sharecrop share; farms tenant share; horses mules pa; horses mules pf; implements machinery pa; ip sh agri 18 65; livestock value pa
1890	ag output value pa; farm value per acre pa; farms owner share; farms sharecrop share; farms tenant share; horses mules pa; horses mules pf; implements machinery pa; livestock value pa
1900	farms owner share; farms sharecrop share; farms tenant share; ip sh agri 18 65
1910	farms owner share; farms sharecrop share; farms tenant share; horses mules pf; ip sh agri 18 65; land improved share
1920	ag output value pa; farm capital pa; farm value per acre pa; farms owner share; farms sharecrop share; farms tenant share; ip sh agri 18 65; labor expenditures total pf; land improved share; land owner share; land tenant share; livestock value pa
1930	crop value total pa; farm capital pa; farm value per acre pa; farms owner share; farms sharecrop share; farms tenant share; implements machinery pa; ip sh agri 18 65; land owner share; land tenant share; livestock value pa; nhgis elec motors pf; nhgis rpt autos share; nhgis rpt electric share; nhgis rpt gasengine share; nhgis rpt tractors share; nhgis rpt trucks share; tractors pf
1940	farm capital pa; farm value per acre pa; farms owner share; farms sharecrop share; farms tenant share; horses mules pa; horses mules pf; implements machinery pa; ip sh agri 18 65; labor hired persons pf; land owner share; land tenant share; livestock value pa; tractors pf
1950	farm capital pa; farm value per acre pa; farms owner share; farms tenant share; horses mules pa; horses mules pf; ip sh agri 18 65; livestock value pa; tractors pf

Notes: This table lists the component indicators that enter the agricultural development sub-index in each decade—those passing the builder’s coverage and nonzero-variance screens (South scope). The indicators are averaged after standardization, signed so higher values denote greater development. The inputs come directly from the generated selection inventory in the replication pipeline.

TABLE 19: Selected Inputs for the Industrial Development Index by Decade

Decade	Selected index inputs (South)
1880	ip sh manuf 18 65; pop density; rr mktaccess
1890	—
1900	ip sh manuf 18 65; ln mfg capital pc; ln mfg estab pc; ln mfg output pc; ln mfg output perworker; ln mfg wages pc; pop density; rr mktaccess; sh mfgchild wagek
1910	ip sh manuf 18 65; pop density
1920	ip sh manuf 18 65; ln mfg estab pc; ln mfg output pc; ln mfg wages pc; ln mfghp pc; ln mfgvalad pc; pop density; rr mktaccess
1930	ip sh manuf 18 65; pop density
1940	ip ln avg incwage 18 65; ip sh manuf 18 65; ln avg incwage all; pop density
1950	ip ln avg incwage 18 65; ip sh manuf 18 65; pop density

Notes: This table lists the component indicators that enter the industrial development sub-index in each decade (South scope). A “—” marks a decade with too few available indicators to construct the index (1890). The indicators are averaged after standardization, signed so higher values denote greater development. The inputs come directly from the generated selection inventory in the replication pipeline.

TABLE 20: Selected Inputs for the Human-Capital Index by Decade

Decade	Selected index inputs (South)
1880	hp pat per100k total; ip avg occscore 18 65; ip lit rate 16p; ip sh school 6 16
1890	hp pat per100k total; sh school all
1900	hp pat per100k total; ip avg occscore 18 65; ip lit rate 16p; ip sh school 6 16
1910	hp pat per100k total; ip avg occscore 18 65; ip lit rate 16p; ip sh school 6 16; lit rate all
1920	hp pat per100k total; ip avg occscore 18 65; ip lit rate 16p; ip sh school 6 16; lit rate all
1930	hp pat per100k total; ip avg occscore 18 65; ip lit rate 16p; ip sh school 6 16
1940	hp pat per100k total; ip avg educ years 25p; ip avg occscore 18 65; ip hsplus share 25p; ip sh school 6 16
1950	hp pat per100k total; ip avg educ years 25p; ip avg occscore 18 65; ip hsplus share 25p; ip sh school 6 16

Notes: This table lists the component indicators that enter the human-capital sub-index in each decade (South scope). The indicators are averaged after standardization, signed so higher values denote greater development. The inputs come directly from the generated selection inventory in the replication pipeline.

TABLE 21: Selected Inputs for the White Human-Capital Index by Decade

Decade	Selected index inputs (South)
1880	hp pat per100k total; ip avg occscore 18 65 w; ip lit rate 16p w; ip sh school 6 16 w
1890	hp pat per100k total; sh school w
1900	hp pat per100k total; ip avg occscore 18 65 w; ip lit rate 16p w; ip sh school 6 16 w
1910	hp pat per100k total; ip avg occscore 18 65 w; ip lit rate 16p w; ip sh school 6 16 w; lit rate w
1920	hp pat per100k total; ip avg occscore 18 65 w; ip lit rate 16p w; ip sh school 6 16 w; lit rate w
1930	hp pat per100k total; ip avg occscore 18 65 w; ip lit rate 16p w; ip sh school 6 16 w
1940	hp pat per100k total; ip avg educ years 25p w; ip avg occscore 18 65 w; ip hsplus share 25p w; ip sh school 6 16 w
1950	hp pat per100k total; ip avg educ years 25p w; ip avg occscore 18 65 w; ip hsplus share 25p w; ip sh school 6 16 w

Notes: This table lists the component indicators that enter the white-only human-capital sub-index in each decade (South scope), restricting schooling, literacy, and occupation inputs to white residents. The inputs come directly from the generated selection inventory in the replication pipeline.

D.6. Instrument

The shift-share instrumental variable is created by combining historical migration shares with population shock data. After merging the two, we calculate predicted Black population increases for each destination county, as well as normalized predictions as percentages of 1870 population.

D.7. Controls

The controls we use come from IPUMS full-count census records ([Ruggles et al., 2024](#)), using the 1870 census for baseline controls.

D.8. Jim Crow Laws

We use Jim Crow laws and the Jim Crow index following [Althoff and Reichardt \(2024\)](#). The index is based on the following five factors: the anti-Black share of race-related laws a state passed until 1950, the state’s number of disenfranchisement devices, the state’s

share of congressional delegates that signed the Southern Manifesto, the racial gap in states' school year lengths, and the year in which a state introduced legislation for minimum teacher pay. More details on its construction can be found in [Althoff and Reichardt \(2024\)](#).

D.9. State Legislative Voting Data

TABLE 22: State Coverage of Digitized Legislative Voting Data

State House	State Senate	Jim Crow Votes
Alabama	Alabama	Alabama
Arkansas	Arkansas	Arkansas
Florida	Florida	Florida
Georgia	Georgia	Georgia
Kentucky	Kentucky	Kentucky
Louisiana	Louisiana	Louisiana
Mississippi	Mississippi	Mississippi
Missouri	Missouri	Missouri
North Carolina	North Carolina	North Carolina
Oklahoma	Oklahoma	Oklahoma
South Carolina	South Carolina	South Carolina
Tennessee	Tennessee	Tennessee
Texas	Texas	Texas
Virginia	Virginia	Virginia

Notes: This table shows the current state coverage of our digitized legislative voting data. States in black indicate data availability for State House, State Senate, or Jim Crow roll-call votes; states in gray indicate no data yet digitized.

D.10. Election Data

Our election data collection, based on ICPSR 00001 ([Inter-university Consortium for Political and Social Research, 1999](#)), encompasses presidential and gubernatorial elections across 15 Southern states from 1870 to 1960 (covering Alabama, Arkansas, Florida, Georgia, Louisiana, Mississippi, North Carolina, South Carolina, Tennessee, Texas, Virginia, Kentucky, West Virginia, Maryland, and Missouri). As our main outcome, we focus on counties' Democratic vote shares, defined as the number of Democratic votes divided by the total number of votes in the county. To create a balanced panel, we follow [Bernini et al. \(2023\)](#) in imputing some missing data for specific state-year combinations.

For presidential elections, we create an 1870 variable by combining 1868 data with 1876 values where 1868 data is missing, since no presidential election was held in 1870. We also test an alternative 1870 predictor using 1856 data, imputing missing 1856 Demo-

cratic vote shares through linear regression based on the relationship between 1856 and 1868 vote shares in counties with data for both years. For all other decades (1880–1960), we construct decade-level presidential vote shares as follows: if a presidential election occurred in the decade year (e.g., 1920), we use that value directly; if not (e.g., 1910), we take the average of the two nearest presidential elections (e.g., 1908 and 1912) to proxy the missing decade-year value. We make two exceptions to this approach due to missing data. For Virginia, we use only the 1888 election to proxy the 1890 presidential vote share because data for 1892 are unavailable. Similarly, for Mississippi, we rely solely on the 1952 election to approximate the 1950 value, as data for 1948 are missing. Additionally, we create a pre-1865 variable, which is a combination of 1856, 1860, and 1864 data to minimize missing values. Based on these data, we also compute each county’s change in Democratic vote shares.

For gubernatorial elections, we interpolate vote shares for decade years by averaging the closest preceding and following election results within each state, since gubernatorial election timing varies by state. When a decade year coincides with an actual election year for a given state, we use the observed values directly. We make several state-specific adjustments to this general approach: we use single election years rather than interpolation for Arkansas 1870 (using 1872), Georgia 1870 (using 1868), Georgia 1890 (using 1892), Georgia 1920 (using 1918), Virginia 1870 (using 1873), Virginia 1890 (using 1893), Virginia 1910 (using 1909), Tennessee 1880 (using 1878), and Texas 1870 (using 1869). For Louisiana 1920, we average 1916 and 1924 values due to missing election data. These adjustments address issues including data availability, absence of Democratic candidates in specific elections, and irregular election timing. As for presidential elections, we create a gubernatorial variable for the pre-1865 period by sequentially filling missing values with available election data from 1865 back to 1852, prioritizing the most recent available election result for each county.

The final dataset is composed of one value of Democratic vote shares for each decade from 1870 to 1960, for both presidential and gubernatorial elections.

D.11. Additional Outcomes

We consider different additional outcomes in our analysis. First, we have the number of lynchings that occurred in a county between 1883 and 1941, from [Seguin and Rigby \(2019\)](#). We also collect data on the second KKK from the Virginia Commonwealth University Libraries’ mapping project ([Kneebone and Torres, 2015](#)).

D.12. Individual-Level Outcomes

We also run our SSIV specification at the individual level, instead of the county level. For this, we use four different outcomes, all based on IPUMS full-count census records ([Ruggles et al., 2024](#)): wealth (combined real and personal property), farm ownership, occupational income scores, and literacy rates.